k-Nearest Neighbor (k-NN) Classifier



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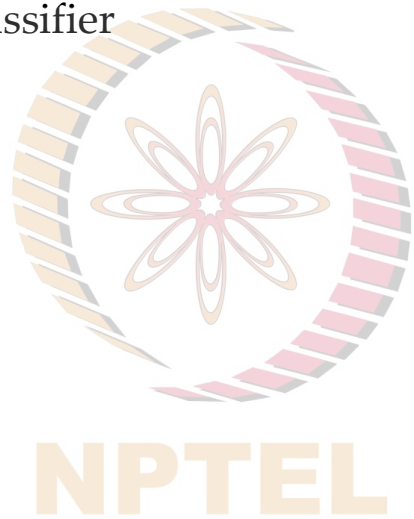
k-Nearest Neighbor (k-NN) Classifier

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Learning Objectives

In this module, we will learn about

- K-Nearest Neighbor (k-NN) classifier
- Advantages
- Limitations
- Example



Non-Parametric ML Model: Revisited

- k-nearest neighbors (k-NN) is a popular non-parametric algorithm that can be used for both regression and classification
- Does not assume a specific functional form for the relationship between the features and target variable
- Estimates the relationship between the features and target variable directly from the training data
- Relatively flexible than parametric ML models
- Can handle complex relationships between the features and target variables, making them particularly useful for high-dimensional and non-linear datasets

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k-NN Classifier: Working

- Based on the **Principle of Proximity**: Similar samples belong to the same class
- Classifies a data sample by considering the class labels of its k nearest neighbors in the feature space
- **Training**: No explicit training is performed in the k-NN Classifier [**Lazy learning algorithm**]
- **Prediction**: The class label is determined by majority voting among the neighbors, where each neighbor's vote is weighted equally; OR it can be *some other chosen criterion of yours*
- Commonly used **distance metrics** include Euclidean distance, Manhattan distance, and Minkowski distance

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Advantages

1. **Simplicity**: Does not require any assumptions about the underlying distribution of the data or any complex mathematical calculations

2. **Versatility**: Can work with both continuous and categorical target variables

3. **Interpretability**: Can provide insight into the relationship between the features and target variable

4. **Efficiency**: No training phase, making it computationally efficient during training

5. **Flexibility**: Can handle multi-class classification and non-linear decision boundaries

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Limitations

1. **Selection of k**: The k value must be chosen carefully.

2. **Sensitivity to outliers and noise**: Relies on the distance between data samples to make predictions. Outliers and noise can adversely affect the predictions

3. **Computationally expensive & Slow**: Can be computationally expensive during predictions for large datasets

4. **Curse of dimensionality**: The distance between data samples becomes less meaningful as the number of features increases.

5. **Class Imbalance**: Can be biased towards the majority class in imbalanced datasets because it tends to predict the class having the majority of samples in the training dataset

Let us Try!

Question: Given the following training data, use KNN classification to predict the target class for a new data sample with x=7 and k=3.

Training Data		
Training Sample Number	**x (feature)**	**y (output/target)**
S1	2	0
S2	5	0
S3	8	1
S4	11	1

You may pause the video and try.

Let us Try!

Answer: Let us calculate the distance of a test data ($x_t = 7$) from each sample of the training dataset

3-Nearest Neighbors

Training Data			
Training Sample Number	**x (feature)**	**y (output/target)**	**Distance of the test sample from all training samples**
S1	2	0	5
S2	5	0	2
S3	8	1	1
S4	11	1	4

By majority voting, the class for $x_t = 7$ is $y = 1$.

To Summarize

In this module, we learned

- K-Nearest Neighbor (k-NN) classifier
- Advantages
- Limitations
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